

Information Retrieval & Data Analysis Project

Project Overview

In this project, students will design and implement a **real-world web intelligence product** that collects, processes, and analyzes web-based textual data and integrates **AI-powered features** to enhance usability and insights.

The goal is to simulate a **functional data product pipeline**, like systems used in industry for:

- market intelligence
- recommendation systems
- content analysis platforms
- decision-support tools

Core Idea

You must build a complete system that:

1. Collects real-world data from the web (scraping + crawling)
2. Stores data in structured format (JSON/XML)
3. Cleans and processes textual data
4. Performs analysis and builds IR-style indexing/search
5. Provides a **working product interface**
6. Integrates at least ONE AI-powered feature inside the system

System Requirements

Every project must include:

1. Web Data Collection
 - Multi-page scraping or crawling
 - Respect robots.txt
 - Minimum dataset size: 50–200 records
2. Data Storage Layer
 - Structured format (JSON or XML)
 - Consistent schema across all records
3. Data Processing Layer
 - Cleaning (tokenization, normalization, stopword removal)
 - Handling missing/noisy data
 - Basic preprocessing pipeline

4. Data Understanding Layer
 - Basic statistical analysis
 - Keyword frequency analysis
 - Dataset exploration
5. Product Layer

Students must build a **functional product**, such as:

 - Search-based system (IR-style retrieval)
 - Insight dashboard (analytics product)
 - Recommendation / grouping system
 - Comparison system (e.g., A vs B analysis)
6. AI Integration Layer

AI must be used as a **feature inside the product**, such as:

 - Sentiment analysis on reviews/news
 - Topic classification (news/job categories)
 - AI-assisted summarization of content
 - Keyword extraction or tagging assistant

Project Goal

To build a **realistic data-driven web product** that transforms raw web content into meaningful, AI-enhanced insights.

Example for Project Domains

1. News Intelligence Products
 - News trend analyzer (politics, sports, tech)
 - Fake vs real news classification support tool
 - News summarization dashboard
2. Product Intelligence Systems
 - E-commerce review analysis system
 - Product comparison tool (Amazon-like analysis)
 - Customer sentiment dashboard
3. Job Market Intelligence
 - Job trend analyzer
 - Skill demand tracking system
 - Job classification and filtering tool
4. Entertainment Analytics
 - Movie review sentiment dashboard
 - Film recommendation explorer
 - IMDb-style analysis tool
5. Food & Recipe Intelligence
 - Recipe recommendation system
 - Ingredient-based search tool
 - Cuisine classification dashboard

6. Travel Intelligence Products
 - Travel destination insights dashboard
 - Hotel review analysis system
 - Tourist trend analyzer
7. Tech Content Platforms
 - Developer article analyzer
 - Programming trend dashboard
 - Tech topic clustering tool

Constraints

- No fully prebuilt datasets (scraping required)
- No black-box AI APIs used as full solution
- Must respect robots.txt
- Minimum dataset size required
- Must include a working product interface (CLI, web app, or dashboard)

